

# MOURYA BIRRU

Portfolio Website - <https://mourya-birru.vercel.app>

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**LeetCode** | **Linkedin** | **GitHub** | **HackerRank**

Completed over 300 LeetCode problems and earned multiple HackerRank certifications in Python, Java, SQL, and problem-solving. MongoDB Certified Associate Developer with hands-on experience in developing full-stack projects using Java, Node.js, Firebase, and MongoDB. Currently interning in the fields of generative AI and cybersecurity. Strong interest in building scalable backend systems, AI/ML-driven applications, and intelligent automation solutions.

## EDUCATION

### Geethanjali College of Engineering and Technology

Computer Science and Engineering Bachelors

CGPA: 8.25

Hyderabad, India

September 2023 - Present

### Resonance Junior College

MPC Intermediate

Percentage: 84.9%

Hyderabad, India

2022 - 2023

### Rajiv Gandhi University of Knowledge Technologies

PUC - 1

Basar, India

2021 - 2022

### Pallavi Model School

Degree in Schooling

Percentage: 87.8%

Hyderabad, India

2015 - 2021

## EXPERIENCE

### Avasan Chakra | Software Engineer Intern

Hybrid | April 2026 - Present

- Performed 12+ simulated penetration testing tasks: scanning, exploitation, and reporting.
- Identified critical vulnerabilities: SQL Injection, XSS, Host Header Injection, IDOR.
- Used tools like Nmap, Wireshark, Metasploit, and Nessus for vulnerability assessment.
- Documented findings and remediation in structured security reports.

### Google | Student Ambassador

July 2025 - December 2025

- Selected as a Student Ambassador for programs led by Google, representing and promoting tech initiatives within the student community.
- Organized and supported workshops, events, and awareness activities to encourage student participation in technology programs.
- Built strong peer networks and increased engagement through campus outreach and online promotion.

## SKILLS

|                        |                                                                                                                                                   |
|------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------|
| Programming Languages: | C, Java, Python, R, HTML, CSS, JavaScript, Node.js, TypeScript                                                                                    |
| Libraries/Frameworks:  | Tensor Flow, PyTorch, Scikit-learn, Keras, OpenCV, NumPy, Pandas, Matplotlib, Seaborn, Statsmodels, React.js, Bootstrap, Tailwind CSS, Express.js |
| Tools / Platforms:     | Git, VS Code, PowerBI, Microsoft Office 365, GitHub, Jupyter Notebook, Google Console, Tableau, Figma, Firebase                                   |
| Databases:             | SQL, MongoDB                                                                                                                                      |

## PROJECTS / OPEN-SOURCE

### Hobby OS (x86 Kernel) | Link

C, x86 Assembly, GNU Assembler, QEMU

A custom-built, low-level x86 operating system kernel developed from scratch.

- Implemented foundational kernel structures including the Global Descriptor Table (GDT), Interrupt Descriptor Table (IDT), and memory paging.
- Architected keyboard driver and hardware-level interface protocols to enable functional user-mode input processing.
- Leveraged Assembly for bootloader bootstrapping and low-level CPU state management, ensuring stable transition to protected mode.

Skills: Systems Programming, Kernel Development, Low-level Optimization, Hardware Architecture

### Air Canvas | [Link](#)

*Python, OpenCV, NumPy, Computer Vision*

An interactive gesture-based painting application using real-time hand tracking.

- Implemented color detection and masking algorithms in OpenCV to track specific colored objects for dynamic drawing inputs.
- Developed frame-by-frame spatial coordinate mapping to translate real-world hand gestures into digital strokes on a virtual canvas.
- Optimized image processing pipelines to minimize latency, ensuring smooth and responsive interaction for a real-time user experience.

Skills: Computer Vision, Real-time Image Processing, Human-Computer Interaction

### Smart Attendance System | [Link](#)

*Python, OpenCV, Computer Vision, Deep Learning, SQL*

A real-time biometric attendance solution designed to automate classroom tracking.

- Implemented facial detection and recognition pipelines using OpenCV and deep learning models to identify students with high accuracy.
- Built a robust registration and training workflow to map unique student identifiers to facial features stored in a relational database.
- Developed an automated logging system that captures timestamps and attendance records, eliminating manual intervention and proxy issues.

Skills: Computer Vision, Real-time Processing, Biometric Authentication, Data Modeling

### Career Compass | [Link](#)

*TypeScript, Firebase, React, Tailwind CSS, Next.js, Genkit*

A full-stack platform for personalized career guidance.

- Implemented Genkit-powered AI for resume analysis, skill mapping, and certification suggestions.
- Built dynamic UI with Tailwind, shadcn/ui, and Radix UI.
- Used Firebase for real-time storage and user authentication.

Skills: Full-stack development, Firebase Auth, AI-integration

### EatIQ | [Link](#)

*TypeScript, N8N, React, Tailwind CSS, Next.js*

Developed EatIQ, a web app that analyzes food images and shows nutrition details using AI.

- Built the frontend using React, TypeScript, and Tailwind CSS with a clean and responsive design.
- Integrated AI workflows to process images and generate calorie and nutrient information automatically.

Skills: React, Node.js, AI integration, n8n, Web Development

## MEMBERSHIPS

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**IEEE Computer Society | Student Member**

September 2024 - January 2026

**Indian Society for Technical Education | Student Member**

February 2024 - Present

## CERTIFICATIONS

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- Networking Technician - **Cisco**
- MongoDB Associate Developer (NodeJS) - **MongoDB**
- Data Science and Machine Learning BootCamp with R - **Udemy**
- Machine Learning in Python - **365 Data Science**
- The Complete Full-Stack Web Development Bootcamp - **Udemy**
- Java - **HackerRank**

## ACHIEVEMENTS

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- Secured an AIR 9602 in GATE 2026 CSE
- Ranked 542 Globally and 149 Nationally in IEEE Extreme 18.0
- Secured Rank 1 in the Smart Interviews GCET-2027-R Coding Challenge among peers across the college